AER1503 Project Definition

Optimizing transfer trajectories for Earth-Mars interplanetary orbital maneuvers

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Abstract

With the emergence of NASA's Artemis program and other global initiatives, the public's interest in space exploration has been renewed. Given that humanity has already been to the Moon (and will likely get there again), the next feasible target for mankind is Mars. Despite its relatively close proximity to Earth, planning Martian missions is notoriously complex. Factors such as continually-changing planet positions, spacecraft design constraints, and mission length all contribute to the difficulty of this mission type. In this project, we seek to address the challenge by developing a comprehensive Python-based software that identifies and optimizes interplanetary transfer trajectories between Earth and Mars. Using the OpenMDAO Python framework, the software will consider several design variables (i.e propulsion methods, transfer times, fuel minimization) in order to recommend optimal missions to Mars.

There are several goals that the project will hopefully meet by the end of the semester. These include: producing an end-to-end Python program that will optimize the transfer trajectory of a Martian mission, providing a comparison of various types of transfer maneuvers (i.e Hohmann, low-thrust, gravity-assisted) and determine what works best for several launch windows, and serving as a starting point for further research in trajectory optimization. These goals may be adjusted as the semester progresses.

Since OpenMDAO is a multidisciplinary optimization (MDO) framework, the project will take advantage of its capabilities by optimizing several subsystems (i.e orbital mechanics, propulsion, thermal, and power) separately, which will then be combined into a larger global environment. For the project to proceed as planned, there are several steps that will be taken. To begin, we will conduct a literature review of OpenMDAO, along with Chapter 10 content from the course AER1503. In particular, an optimal Hohmann transfer tutorial from OpenMDAO will be followed. Then, we will determine exact design variables and decide on spacecraft parameters. These will serve as the basis for what the program will seek to optimize. Following this, we will create a basic architecture of OpenMDAO systems and subsystems, which will be connected together as OpenMDAO inputs and outputs. Results will be presented in a final report that summarizes the project at large, along with conclusions drawn from the results of the transfer trajectory optimization. The aforementioned steps form a general guideline for what the project is expected to cover over the course of 4 months. As the project's scope and realistic outcomes become better defined, these steps may change.

Successful completion of the project as outlined will allow for a deeper understanding of what it will take to send humans to Mars, and might also provide insights that go beyond what is covered in the AER1503 course.